

Day 2 Problem Set

Geometry Summer Credit | Slope, Lines, and Transversals

Name:	Date:	Period:
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Goal

Practice slope, parallel/perpendicular line relationships, and angle equations formed by transversals. Show enough work to make your setup clear.

Support Practice: Build the Pieces

1.	Find the slope for each ratio: rise 6, run 3; rise -8, run 4; rise 5, run -10.
2.	Solve $2x + 14 = 74$.
3.	Solve $5x - 12 = 2x + 33$.
4.	Find the angle supplementary to 118 degrees.

Core Practice: Slope and Line Relationships

5.	Find the slope through $(-3, 4)$ and $(5, 10)$.
6.	Find the slope through $(2, -6)$ and $(2, 7)$. What kind of line is this?
7.	Find the slope through $(-4, 1)$ and $(8, 1)$. What kind of line is this?

8. A line has slope $\frac{2}{5}$. Give the slope of a parallel line and a perpendicular line.

9. A line has slope -3 . Give the slope of a parallel line and a perpendicular line.

10. Are the lines through A(0, 1), B(4, 7) and C(-2, 5), D(2, 11) parallel, perpendicular, or neither?

11. Are the lines with slopes $-\frac{2}{3}$ and $\frac{3}{2}$ parallel, perpendicular, or neither? Explain.

Day 2 Problem Set

Continued

Core Practice: Transversals and Angle Equations

12. A corresponding angle to 82 degrees is labeled $2x + 16$. Find x .

13. Alternate interior angles are labeled $6x - 7$ and $3x + 35$. Find x and the angle measure.

14. Alternate exterior angles are labeled $4x + 9$ and $7x - 30$. Find x and the angle measure.

15. Same-side interior angles are labeled $5x + 20$ and $3x$. Find x and both angle measures.

16. Same-side interior angles are labeled $2x + 15$ and $4x + 45$. Find x and both angle measures.

17. Error analysis: A student says corresponding angles are supplementary. Explain the mistake and give a correct statement.

Mixed Review and Reasoning

18. Spiral: Find the midpoint of $(-8, 3)$ and $(6, -5)$.

19. Spiral: Find the distance between $(0, 0)$ and $(9, 12)$.

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| 20. | Create your own transversal equation problem where $x = 12$. State the angle relationship and solve it. |
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| 21. | Challenge: A quadrilateral has consecutive side slopes 2, $-1/2$, 2, and $-1/2$. What kind of quadrilateral might it be? Explain using slope relationships. |
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Before You Submit

Circle one slope problem and one angle problem you feel confident about. Star one problem you want help with.